Roll Number: _____ Group: _____

Thapar Institute of Engineering & Technology, Patiala
(Deemed to be University)
Department of Computer Science and Engineering
End Semester Test (EST)

Semester: IV **Course Code: UTA018**
Date- 12.05.2026, 09:00 AM **Course Name : Object Oriented Programming**
Time: 03 Hours, M.M.: 40 **Name of Instructor: Dr. Saurabh Arora, Dr. Harpreet**

NOTE: * Attempt any five questions.
** Assume any missing information.
***Solve Questions in order, attempt new problem from new page.

Q.	Questions	Marks	CO	BL
Q1.	**a)** Explain the four pillars of Object-Oriented Programming. Define each concept with a C++-specific explanation. **b)** Discuss the concept of constructor overloading with example. What is the role of destructor in dynamic memory management.	[4+4]	CO1	L2
Q2.	A hospital is developing a software system to manage doctors, surgeons, and AI-assisted robotic surgeons. The following conditions are given: Class MedicalStaff contains common details like staffID and department. Class Doctor and class Researcher inherit from MedicalStaff. Class RoboticSurgeon inherits from both Doctor and Researcher. Function performDuty() behaves differently for each derived class. The system stores objects using pointers of type MedicalStaff. The hospital wants to ensure that every medical entity implements its own version of generateReport(). **a)**Differentiate between Compile-time Polymorphism and Run-time polymorphism. Give one suitable example of each from the above scenario. **b)** Why should performDuty() be declared as a virtual function? What will happen if it is not declared virtual? **c)** The class MedicalStaff contains a pure virtual function: virtual void generateReport() = 0; Answer the following: What type of class is MedicalStaff now? Can objects of this class be created? Why are pure virtual functions useful in real-life systems? **d)** Suppose the destructor of MedicalStaff is not virtual and objects are deleted using base class pointers. Explain the possible problem that may occur during object destruction in RoboticSurgeon	[8]	CO2	L4
Q3.	What are the different STL components. Expalin the deque container with following operations: Insert at begibing, Insert at end, Remove from beginning, Remove from end , Access first and last element, Clearing all elements.	[8]	CO4	L3
Q4.	**a)**Discuss the concept of Operator Overlaoding as class members and as friend functions with example. Which operators cannot be overloaded in both cases. **b)** Write a C++ program to overload Comma operator.	[4+4]	CO3	L4/L6

Q5.	Design a templated class LibraryShelf, that represents a Shelf Management System in a Smart Library, with information about books stored on a shelf. The goal is to allow flexibility in how books are represented—either by book titles (string) or accession numbers (int).
	The LibraryShelf class contains the following data members: T books[30] : An array of books (templated as T) with a maximum size of 30
	int bookCount :Tracks the current number of books on the shelf
	int totalIssued: Total number of books issued
	int totalReturned :Total number of books return
	Below is the partial code of the LibraryShelf class template. Complete the code by performing the following tasks:
	I. Implement the default constructor and parameterized constructor. Ensure the parameterized constructor properly initializes the books array.
	II. Implement the functions addBook() and removeBook().
	Books should be added only if the shelf has space (maximum 30 books).
	Books should be removed only if they exist on the shelf.
	template <class T>
	class LibraryShelf
	{
	private:
	T books[30];
	int bookCount;
	int totalIssued;
	int totalReturned;
	
	public:
	LibraryShelf() { }
	LibraryShelf(T bookArray[], int count, int issued, int returned){}
	void addBook(T book){}
	void removeBook(T book){}
	};
	EX1: Input – int books[] = {101, 205, 309};
	LibraryShelf<int> shelf1(books, 3, 12, 9);
	shelf1.addBook(412);
	shelf1.removeBook(205);
	Output :
	Books: 101 309 412
	Total Issued: 12
	Total Returned: 9
	EX2: Input – string books[] = {"Physics", "Chemistry", "Mathematics"};
	LibraryShelf<string> shelf2(books, 3, 20, 18);
	shelf2.addBook("Biology");
	shelf2.removeBook("Chemistry");
	Output :
	Books: Physics Mathematics Biology
		[8]	CO4	L6